

```
### Základní fce vektoru
```

```
a <- 1:10
```

```
mode(a)
```

```
length(a)
```

```
puk <- 21:30
```

```
b <- c(27, 20, 35)
```

```
names(b) <- c("humerus", "radius", "femur")
```

```
b
```

```
names(b)
```

```
as.numeric()
```

```
as.character()
```

```
nope <- c(0,1,0,0,0,0,1,1,1)
```

```
as.logical(nope)
```

```
nope2 <- c(FALSE, FALSE, TRUE, TRUE, TRUE, FALSE,  
TRUE, TRUE, TRUE)
```

```
as.numeric(nope2)
```

```
a <- 1:9
```

```
nope2*a
```

```
### Vyber prvku vektoru
```

```
x2 <- letters; x2
```

```
x2[5]
```

```
x2[1:5]
```

```
x2[c(1,5)]
```

```
head(x2)
```

```
head(x2,3)
```

```
tail(x2)
```

```
tail(x2,4)
```

```
x3 <- 1:20;x3
```

```
x3[x3<13 & x3>5]
```

```
b["humerus"]
```

Operace s vektory

Pridani noveho prvku, odstraneni prvku a
preskladani

```
y <- 1:7  
y[6] <- 7; y  
y <- c(y[1:2],70,y[3:7]); y  
y[9] <- 15; y  
y[13] <- 13; y  
y <- y[!is.na(y)]; y
```

```
summary(b)  
str(b)
```

```
a <- c(1,43,86,12,57,98)  
b <- c(72,54,82,22,1,23)  
c <- c(55, 13)  
d <- c("a","b")
```

```
c(a, c)                # spoji vektory do delsiho  
vektoru  
cbind(a, b)             # spoji vektory (i matice)  
jako sloupce (nutny stejny pocet radku matic)  
rbind(a, b)             # spoji vektory (i matice)  
jako radky (nutny stejny pocet sloupcu matic)  
paste(a, b, c, sep="") # spoji vse do textu  
cbind(a, c)  
rbind(a, c)
```

Vyzkousejte:

```
a+b  
a*b  
a+c  
a*c
```

```
a<c
```

```
a-c
```

```
a-b
```

```
d <- c(FALSE, TRUE, TRUE, FALSE, FALSE, TRUE)
```

```
a*d
```

```
a[7] <- 25
```

```
cbind(a, c)
```

```
### Vektor jako retezec textu
```

```
k <- "Coffee is your only hope"
```

```
l <- c("Coffee", "is", "your", "only", "hope")
```

```
nchar(k)
```

```
nchar(l)
```

```
sub("Coffee", "Caffeine", k)
```

```
paste(l, collapse = ' ')
```

```
strsplit(k, ' ')
```

```
substring(l, 1, 1)
```

```
##### FAKTOR
```

```
#####
```

```
f <- c("humerus", "ulna", "ulna", "radius",
```

```
"femur", "ulna", NA)
```

```
as.factor(f)
```

```
# zmena poradi levels faktoru
```

```
# note: NA se automaticky do levels nezarazuje
```

```
factor(c("humerus", "ulna", "ulna", "radius",
```

```
"femur", "ulna", NA), levels =
```

```
c("humerus", "radius", "ulna", "femur"))
```

```
factor(c(1, 1, 2, 1, 3, 4, 4, 2, 1))
```

```
factor(c(1, 1, 2, 1, 3, 4, 4, 2, 1), levels=4:1)
```

```
# zmena poradi levels
```

```
factor(c(1, 1, 2, 1, 3, 4, 4, 2, 1),
```

```
levels=4:1,labels=c("a","b","c","d")) # jiné názvy
levels
factor(c(1,1,2,1,3,4,4,2,1), exclude=1)
# nezahrnutí 1 (1 převedena na NA hodnoty)
```

```
f2 <- as.factor(1:5); f2
mean(f2)
mean(as.numeric(f2))
as.factor(f2)
as.numeric(f2)
as.character(f2)
as.matrix(f2)
as.list(f2)
as.data.frame(f2)
```

```
##### MATICE
```

```
#####
matrix(NA, nrow=2, ncol=2)
```

```
m <- 1:20;
matrix(m, 4, 5)
matrix(m, ncol=5)
matrix(m, nrow=4)
m <- matrix(m, 4)
matrix(m, c(4, 5))
dim(m) <- c(4, 5);m
matrix(m, byrow=TRUE)
```

```
m_2 <- 1:10
POK <- matrix(m_2, 4, 5, byrow = TRUE)
dim(POK)
vek <- 1:40
POK
```

```
### Vyber z matice - podmatice
m
```

```
m[1,2]
m[1:3,]
m[,2]
m[c(1,3),]
# vypsaní druhého sloupce jako "sloupcový vektor"
m[,2, drop=FALSE]
```

```
## funkce matic
```

```
nrow(m)      # počet radku matice m
ncol(m)      # počet sloupce matice m
diag(m)      # diagonála matice m
rowSums(m)   # sumarizační funkce radku (vrátí
součet hodnot v každém radku)
colSums(m)   # sumarizační funkce sloupce (vrátí
součet hodnot v každém sloupci)
dimnames(m)  # názvy sloupce a radku matice
rownames(m)  # názvy radku matice
colnames(m)  # názvy sloupce matice
```

```
# příklady přiřazení názvu radkům a sloupcům
matice
```

```
rownames(m) <-
c("ID_001", "ID_002", "ID_003", "ID_004")
colnames(m) <- c("femur F1", "humerus H1", "ulna
U1", "radius R1", "tibia T1")
```

```
A <- matrix(1:20, ncol = 5); A
B <- matrix(letters[1:20], ncol = 5); B
```

```
cbind(A,B)
rbind(A,B)
mode(rbind(A,B))
typeof(rbind(A,B))
```